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COMPUTER SCIENCE **0478/22**

Paper 2 Algorithms, Programming and Logic **February/March 2025**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **16** pages. Any blank pages are indicated.

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10 A sports club uses a six-character alphanumeric membership code to identify each member of the club.

The one-dimensional (1D) array `MemberID[]` is used to store the unique membership codes for club members.

The two-dimensional (2D) array `Name[]` is used to store the names of the club members. The first and last name of each member will be stored in separate array elements.

The system can store details for a maximum of 1000 members.

The position of any member's data is the same in both arrays. For example, the data in index 2 of `MemberID[]` belongs to the member in index 2 of `Name[]`

The variable `NewID` is used to input a new membership code.

Write a program that meets the following requirements:

- Provide a menu that offers the choices: inputting a new member, outputting a list of membership codes and first and last names, or stopping.
- Input and validate a response to the menu.
- When inputting a new member, input a new membership code and check that it contains six characters:
  - If the new code is six characters, check it against all the previously stored membership codes to make sure it is unique.
  - If the code is **not** unique, a new code must be entered and checked.
  - If the code is unique, it is stored in the first available space in the appropriate array and the new member is required to enter their first name and last name, which are also stored in the corresponding location of the appropriate array.
- When outputting a list of membership codes and names, output for each member: their membership code, first name and last name.
- The program will continue until the stop option on the menu is selected.

You must use pseudocode or program code **and** add comments to explain how your code works.

You do **not** need to declare any arrays, variables or constants; assume this has already been done.

You do **not** need to initialise the data in the arrays.

You do need to initialise any variables or constants used if appropriate.

All inputs and outputs must contain suitable messages.

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Error 2 line number .....

Correction .....

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Error 3 line number .....

Correction .....

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Error 4 line number .....

Correction .....

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Error 5 line number .....

Correction .....

..... [5]

(b) Write the pseudocode statement to output the average of the numbers, rounded to **two** decimal places.

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..... [2]

(c) Explain how you could change the corrected algorithm so that it also finds the smallest number that was input and outputs its value at the end.

Any code used must be fully explained.

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..... [4]

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9 Consider this logic expression:

Z = (**A** NAND **B**) XOR (NOT (NOT **B** NAND **C**))

(a) Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify the logic expression.

[5]

(b) Complete the truth table from the given logic expression.

A	B	C	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

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(a) Complete the trace table for the input data:

Y, 7, S, 9, Y, 5, M, 12, Y, 25, D, 5, N, 10, A, 6

Answer	Value1	Operator	Value2	OUTPUT

[5]

(b) State the purpose of the algorithm on **page 6**.

..... [1]

(c) There are problems with the given algorithm. If the lower-case letter 'y' is entered for `Answer`, the algorithm stops and if a lower-case letter is entered for `Operator`, the algorithm gives an incorrect output.

(i) Explain how you would change the algorithm to prevent the problems described.

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..... [3]

(ii) Identify **one** more error related to inputs when the algorithm is run.

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..... [1]